

PHYVER: Physics-Grounded Material Claim Verification with Multi-Fidelity Physical Evidence

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Demo Link: <http://zhoulab-1.cs.vt.edu:5557/demo>

Abstract

Material claims in papers, patents, *etc.*, often involve physical feasibility (*e.g.*, stability under conditions, property consistency), not just textual feasibility. Yet most claim verifiers operate over language, therefore producing ungrounded judgments. On the other hand, direct first-principles verification (*e.g.*, density functional theory, DFT) is inflexible and hard to invoke from underspecified free-form claims. Therefore, we introduce PHYVER, a physics-grounded material claim verification system that bridges this gap by translating claims into multi-fidelity physical evidence and interpretable verdicts. To support human-in-the-loop inspection, we present an interactive web interface that visualizes the instantiated structure, optimization trajectories, DFT summaries, and the final decision. On expert-labeled claims, PHYVER improves agreement with experts over text-only GPT-5.1, reducing MAE from 1.54 to 1.20 and Signed MAE from 0.95 to 0.82, and increasing Accuracy@ ± 1 from 50% to 70%.

1 Introduction

Material-related claims appear throughout scientific papers, patents, technical reports, and even everyday product descriptions (Tshitoyan et al., 2019; Smith et al., 2022). Unlike purely textual factuality, many material claims are related to whether a proposed material configuration is physically feasible and whether its implied properties are consistent with underlying thermodynamic and structural constraints (Evans et al., 2021; Jain et al., 2013). As large language models (LLMs) become a popular interface for scientific reading and discovery, an increasingly practical question emerges: *given a free-form material claim, can we verify it with evidence that is grounded in physics rather than linguistic plausibility?*

Existing claim verification pipelines in natural language processing (NLP) predominantly operate over text: they retrieve and assess evidence sentences, or rely on an LLM’s internal priors (Thorne et al., 2018a; Wadden et al., 2020a; Manakul et al., 2023). However, for materials, the core evidence for many material claims is not another sentence in a corpus, but quantitative physical signals, such as whether a structure can relax to a low-energy configuration, whether computed properties align with the claim, and whether the claim remains consistent under reasonable assumptions (Jain et al., 2013). A claim may be commonly stated yet physically implausible; conversely, a novel but valid claim may be absent from the literature and therefore hard to “verify” by retrieval. This mismatch makes text-only verification brittle and encourages failure modes where an LLM produces fluent but ungrounded rationales.

A seemingly natural alternative is to verify material claims by appealing directly to first-principles simulations such as density functional theory (DFT). Yet DFT is not a verifier by itself: it requires a well-defined atomic structure, careful choices of computational settings, and most importantly, a mapping from natural-language claims to executable tests and measurable proxies (Lejaeghere et al., 2016). In practice, the bottleneck is often *knowledge and specification*: translating an underspecified claim into a concrete material candidate (formula and atomic positions), deciding what to compute, and interpreting results against the claim’s semantics. As a result, physics tools are powerful but difficult to invoke reliably from natural language, while NLP tools are accessible but lack intrinsic physics awareness. This leaves a clear gap for methods that can translate free-form claims into well-specified simulations and return verdicts grounded in quantitative physical evidence.

We address this gap with **physics-grounded**

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⁺A demo video can be accessed at [This Link](http://zhoulab-1.cs.vt.edu:5557/demo).

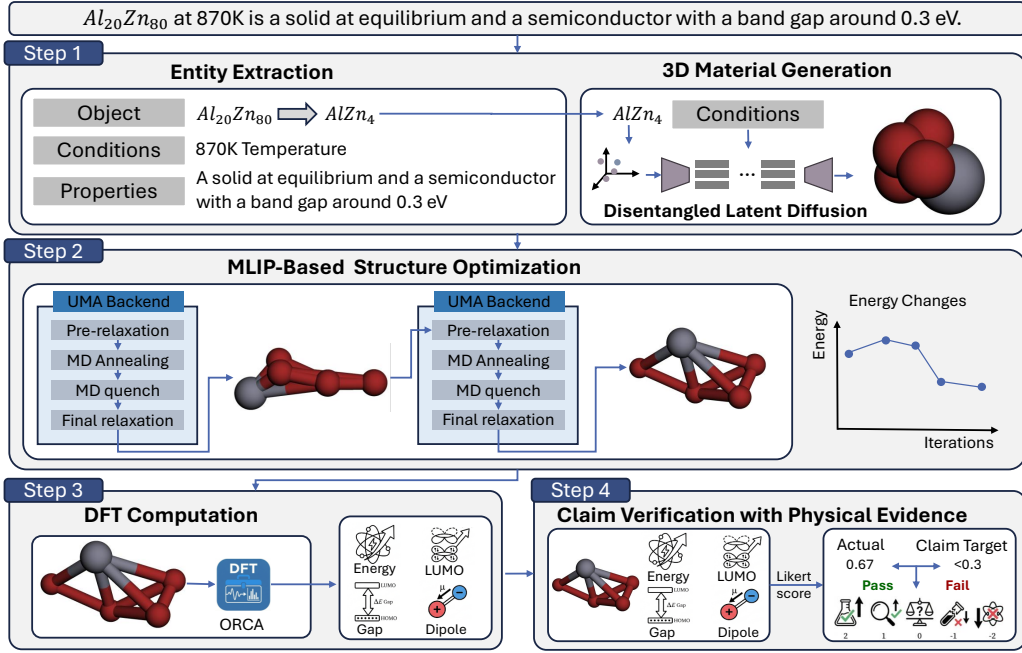


Figure 1: PHYVER system for claim verification contains four steps: Step 1. material instantiation, Step 2. MLIP-based optimization, Step 3. DFT computation, and Step 4. evidence-grounded claim verification.

material claim **verification** (PHYVER system), an end-to-end system that makes claim verification both *knowledge-aware* and *physics-grounded*. Given a textual claim, PHYVER at first instantiates material candidates from text, then gathers physical evidence through multi-fidelity simulation (low-fidelity machine-learned interatomic potentials, MLIP, and high-fidelity DFT computation), and finally produces an interpretable verdict explicitly supported by the multi-fidelity evidence. For material instantiation, we introduce a text-to-structure module, which translates free-form descriptions into material structures, including chemical formula and atomic positions. Physical evidence is then acquired efficiently via a multi-fidelity workflow: a fast optimization stage based on MLIP performs iterative structural relaxation toward energy minimization, producing an optimization trajectory and energy-change profile; for higher-confidence confirmation, we selectively invoke DFT to compute key properties needed to evaluate the claim. By grounding verification in physical evidence, the system reduces reliance on linguistic priors and enables judgments that an LLM alone cannot reliably infer.

To demonstrate PHYVER system, we present an interactive demo designed for human-in-the-loop scientific use. Users input a textual claim and receive a structured, inspectable verification report

comprising a 3D visualization of the instantiated material structure, the energy evolution during structural optimization, computed DFT properties, and a final verdict with five graded outcomes (*true*, *possibly true*, *uncertain*, *possibly false*, *false*) according to Likert scale (Likert, 1932). Crucially, the interface exposes intermediate artifacts and evidence summaries, allowing users to understand *why* a claim is supported or contradicted, and to iteratively refine ambiguous claims rather than treating verification as a black-box decision.

Our contributions are threefold:

- **System:** We introduce a physics-grounded claim verification pipeline that connects natural-language material claims to executable, evidence-producing workflows, and aggregates multi-fidelity physical evidence (MLIP optimization and DFT) into evidence-grounded verdicts.
- **Demo:** We build an interactive interface that visualizes instantiated structures, optimization trajectories, computed properties, and supporting evidence to enable transparent claim verification.
- **Evaluation:** We provide empirical results on expert-rated material claims, demonstrating improved alignment with expert judgments compared to text-only LLM assessment.

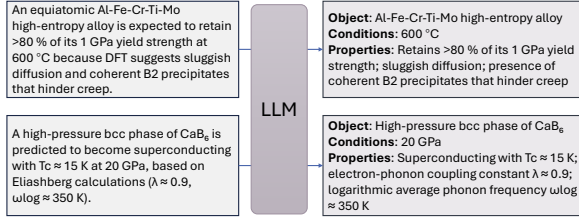


Figure 2: Two example results of entity extraction.

2 System Design

LLMs can assess a claim using their internal knowledge, but they struggle when a claim expresses a novel hypothesis or requires physical feasibility beyond what is stated in text. For materials, whether a claim is true often depends on evidence from the 3D physical world, *e.g.*, whether a proposed structure can relax to a stable configuration under realistic conditions. Motivated by this gap, our system performs verification through four steps as shown in Figure 1: Step 1. **3D material instantiation**, which maps a claim to a plausible candidate structure; Step 2. **MLIP-based material optimization** which runs molecular dynamics (MD) to obtain a stable 3D structure and optimization evidence; Step 3. **DFT computation**, which selectively computes claim-relevant physical indicators via a developed DFT toolbox; and Step 4. **evidence-grounded claim verification**, which aggregates the computed evidence to produce a knowledge-aware and physics-grounded verdict and reasons.

2.1 Material Instantiation

Entity Extraction. A free-form material claim is often underspecified for computation: it may mix composition, operating conditions, and target properties in a single sentence, which makes it difficult to directly instantiate a concrete structure and verify it against physical evidence. To make the claim executable, we first decompose it into a small set of structured entities: (i) the material object (*e.g.*, chemical formula, dopants, or prototype keywords), (ii) conditions (*e.g.*, temperature, pressure, or environment), and (iii) claimed properties (such as stability, phase, or electronic/mechanical descriptors). This structured representation serves as the interface between language and physics: it determines what candidate structures to generate and what properties to compute and compare. In practice, we find that LLMs are reliable at extracting such entities, especially in the molecular do-

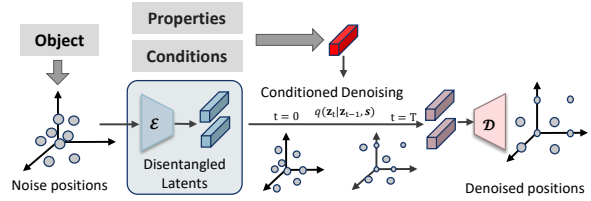


Figure 3: Disentangled latent diffusion model.

main (Jung et al., 2024). Therefore, we implement entity extraction by prompting an LLM to return a structured output for the following modules. Figure 2 shows two entity extraction examples.

3D Material Generation. By moving forward from previous work (Chen et al., 2025), we introduce the disentangled latent diffusion framework to generate a plausible 3D material candidate conditioned on the extracted entities. As illustrated in Figure 3, given the extracted material object (composition) and optional property/condition constraints, the generator first assigns atom types according to the composition and initializes their 3D coordinates by sampling noisy (Gaussian) positions. An encoder then maps this noisy structure into a disentangled latent space consisting of node-wise latents that capture local geometric patterns and a global property latent that summarizes property-relevant semantics. In this latent space, the model performs an iterative denoising diffusion process conditioned on the property and condition entities, gradually steering the sample toward structures that satisfy the claim specification. Finally, a decoder transforms the denoised latents back into an explicit 3D geometry (atomic positions and lattice), which is used as the instantiated candidate for physical optimization.

2.2 MLIP-Based Material Optimization

The structure produced by the previous step is a hypothesis. It is often noisy, metastable, or even physically inconsistent (*e.g.*, unrealistically short bonds) because it is created to match language constraints rather than to minimize energy. Running DFT directly on such a raw candidate is brittle and misleading. We therefore introduce an optimization stage: an MLIP can approximate the potential-energy surface at orders-of-magnitude lower cost than first-principles methods, allowing us to repair generated structures into a stable local minimum. This process produces MLIP-level fidelity evidence (energy/forces) that is directly useful for

claim verification (Faiyad and Martini, 2025).

We implement this stage with UMA model (Wood et al., 2025) as the default MLIP backend, wrapped through ASE (Hjorth Larsen et al., 2017) to standardize optimization, MD, and trajectory logging for inspection in the demo. As shown in Step 2 of Figure 1, because a single relaxation can be trapped in a poor local minimum (especially for generated candidates), we adopt an iterative workflow, *i.e.*, $pre-relax \rightarrow anneal \rightarrow quench \rightarrow re-relax$. Concretely, we repeat an $anneal \rightarrow quench \rightarrow relax$ loop, *i.e.*, heating to escape unfavorable basins, cooling to 0 K to stabilize, and performing 0 K optimization to refine the geometry, while retaining the best-energy structure across loops.

This stage outputs more than a “final structure”, and it produces physical evidence that later supports the verdict: (i) the optimized structure (for visualization and as DFT input), (ii) the optimization trajectory (to show whether relaxation was smooth or unstable), and (iii) an energy evolution profile (to quantify whether the candidate consistently moves toward a stable low-energy basin). In the web interface, these are directly rendered as the optimized 3D structure plus an energy/metric plot over time, enabling users to interpret the results. As a demo intended for interactive use, we expose different presets (*e.g.*, quick/standard/thorough) that adjust the number of loops, peak anneal temperature, cooling schedule, and force thresholds. This lets users quickly screen claims and tune the simulation settings.

2.3 DFT Computation

While MLIP optimization provides low-fidelity evidence, some claims require higher-confidence confirmation or property evaluation. We therefore develop a DFT (Kohn and Sham, 1965) toolbox for selective, higher-fidelity evidence. Our DFT toolbox follows two principles: (i) it bridges fixed, reproducible DFT settings with flexible, claim-specific property queries, (ii) it runs selectively as a higher-fidelity check beyond MLIP evidence. Concretely, we use ORCA as the backend and run single-point energy and analytical gradients (*e.g.*, SP and EnGrad) on a non-periodic copy of the relaxed structure (Neese, 2012, 2022).

The demo exposes both the raw ORCA input/output and the parsed summary, so users can audit the evidence behind a verdict rather than treating DFT as a black box.

2.4 Evidence-Grounded Claim Verification

To avoid text-only verification driven by linguistic plausibility, we ground the final judgment in physical evidence produced by the MLIP optimization and (optionally) DFT computation. Given the extracted entities, we translate the claim into a small set of checkable predicates (feasibility/stability and property constraints, *etc.*), which resolves the specification gap between free-form language and executable physics tests. We then normalize the simulation outputs into a structured evidence record, including MLIP optimization logs (*e.g.*, energy decrease, final stability indicators, convergence flags) and DFT properties (*e.g.*, dipole, HOMO/LUMO and gap), and compute an evidence-grounded confidence score by comparing each obtained property with the target in the claim. The evaluation follows the Likert scale standard (Likert, 1932). Specifically, the system outputs a five-graded verdict (*true, possibly true, uncertain, possibly false, false*) together with concise reasons (Manakul et al., 2023). In the demo, the evidence is presented as linked panels, a *3D structure viewer* (instantiated/relaxed geometry), an *optimization energy curve* (MLIP trajectory), and a *DFT property table* (parsed outputs), so users can directly inspect what supports or contradicts the claim.

3 Interactive Interface

PHYVER provides a web-based interactive interface that enables step-wise execution and transparent inspection of the full claim verification pipeline. As shown in Figure 4, the interface is organized into two modules: a **control module** (left panel) and a **visualization module** (right panel).

Control Module. The left panel allows users to configure and execute each stage of the pipeline. Users can input a free-form material claim, select the LLM model, choose execution presets (*e.g.*, quick/sprint/standard/thorough), configure hardware and DFT parameters (*e.g.*, device, ORCA cores, memory), and optionally enable or disable DFT. The system supports both step-wise execution ($Generate \rightarrow Optimize \rightarrow DFT \rightarrow Verify$) and one-click end-to-end runs. Each step can be triggered independently, and the current job state, logs, and runtime metadata are displayed in real time. This design enables flexible experimentation, ablation testing, and interactive debugging.

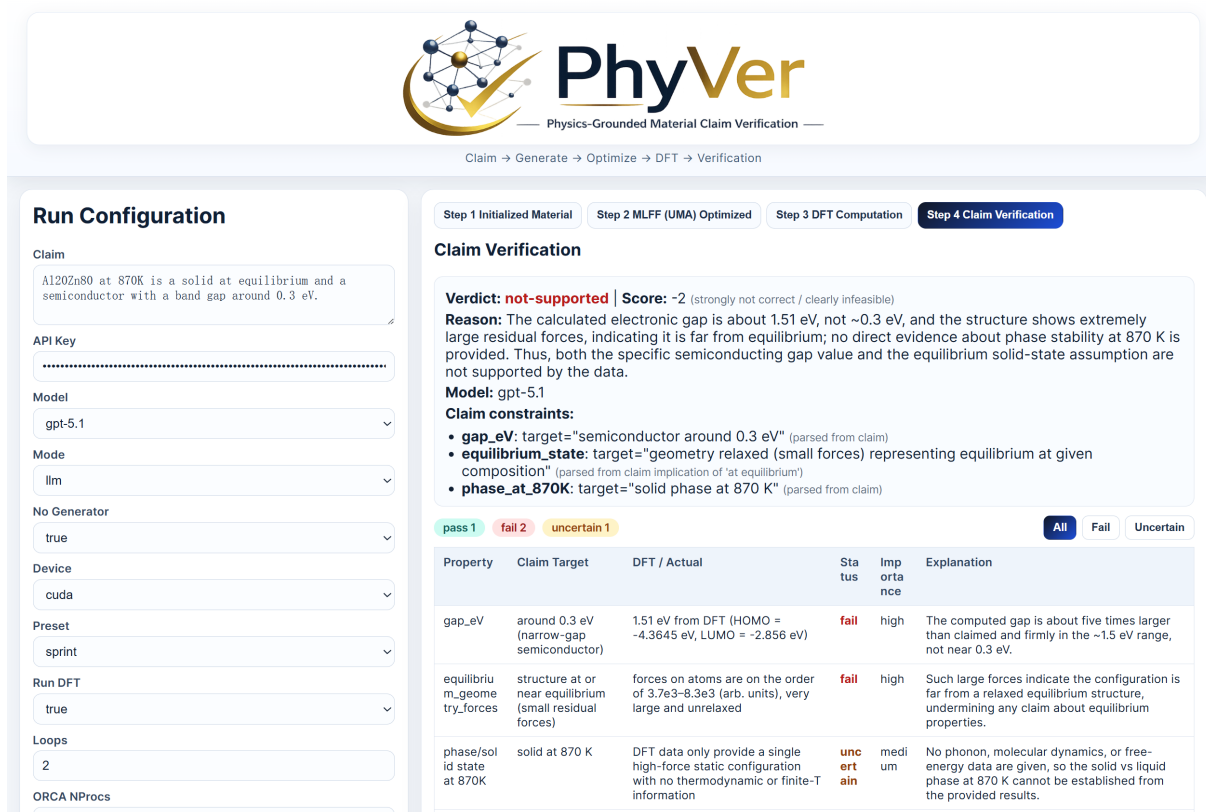


Figure 4: User interactive interface page view.

Visualization Module. The right panel presents structured outputs corresponding to each pipeline stage. For **Step 1**, the initialized 3D structure is rendered using an interactive molecular viewer, together with parsed structure metadata. For **Step 2**, the MLIP-optimized structure is shown alongside quantitative optimization evidence, including energy evolution, volume changes, molecular root-mean-square deviation, and trajectory logs. For **Step 3**, DFT results are summarized as structured evidence cards (e.g., total energy, forces, dipole, HOMO/LUMO, gap), with raw output accessible for transparency. Finally, **Step 4** displays the verification summary, including verdict, confidence score, constraint-level checks, and explanation, together with the raw JSON response returned by the verification module.

4 Experiments

Case Study. To demonstrate the practicability and effectiveness of PHYVER, we conduct two text reasonable while physically infeasible case studies via model o4-mini (Hurst et al., 2024). As shown in Figure 5, the case study presents a failing example where PHYVER evaluates a claim by instantiating a structure, running UMA relax-

ation and DFT, and concluding the claim is not supported due to unstable geometry and failed evidence.

From this case study, we have the following observations: **First**, entity extraction is essential to make an underspecified claim executable, decomposing it into *object/conditions/properties* directly determines what structure to build and what checks to run. **Second**, the MLIP optimization stage provides strong feasibility evidence: large residual forces after relaxation offer a physics-grounded signal that the instantiated interstitial configuration is far from equilibrium, supporting a negative verdict even before relying on textual priors. **Third**, property-level claims that are inherently *comparative* (interstitial vs. substitutional) require matched baselines and consistent reference energies; without an explicitly computed substitutional counterpart, the system can correctly flag the claim as fail, highlighting where additional computations are needed for a verification. The other study can be found in Appendix.

Quantitative Results. We compare PHYVER-GPT-5.1 with vanilla GPT-5.1 on expert-created material claims, each labeled with a 5-point Lik-

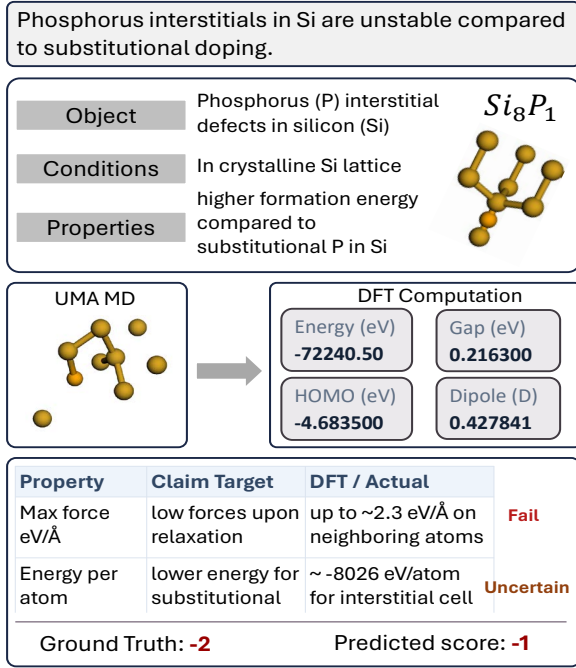


Figure 5: Case study for claim verification. The predicted score and gold score are correctly matched.

Metric	GPT-5.1	PHYVER
MAE ↓	1.5417	1.2000
Accuracy@±1 ↑	0.5000	0.7000
Signed MAE ↓	0.9545	0.8182

Table 1: Comparison between GPT-5.1 (Singh et al., 2025) and PHYVER on expert-labeled material claims.

ert gold score in $\{-2, -1, 0, 1, 2\}$. We report MAE, Accuracy@±1 (within one Likert level), and Signed MAE (directional bias), with implementation details deferred to the appendix. As shown in Table 1, PHYVER improves all metrics: MAE decreases from 1.54 to 1.20 (22.2% relative reduction) and Accuracy@±1 increases from 50% to 70% (+20 points), indicating more reliable decisions when grounding verification in physical evidence. Signed MAE also drops from 0.95 to 0.82, suggesting reduced systematic bias in the predicted direction.

5 Related Works

LLMs for Claim Verification and Scientific Tool Use. Prior work on claim verification in NLP largely relies on *textual evidence retrieval* followed by entailment-style judgment, ranging from general-domain settings (e.g., FEVER) to scientific variants (e.g., SciFact) (Thorne et al., 2018b; Wadden et al., 2020b). While effective when

truth is encoded in the literature, such pipelines do not test *physical feasibility*, which is essential for many material claims. In parallel, *tool-augmented LLM* frameworks connect language models to external solvers or labs (e.g., chemistry toolkits or robotic experimentation) to extend reasoning beyond text (Boiko et al., 2023; M. Bran et al., 2024). Our work follows this direction but differs in purpose: we use the LLM primarily as a *translator* from free-form claims to executable tests, while the final decision is grounded in computed physical evidence rather than the LLMs linguistic priors (Zhong et al., 2024).

Multi-Fidelity Computational Materials Workflows. On the physics side, established infrastructures enable standardized, large-scale *forward* property evaluation from a given structure, including workflow managers and databases built around DFT screening (Jain et al., 2013; Mathew et al., 2017; Huber et al., 2020). More recently, MLIPs have enabled fast structural relaxation and molecular dynamics at near-DFT fidelity, making *multi-fidelity* pipelines (MLIP screening → selective DFT refinement) increasingly common (Battaglia et al., 2022; Zuo et al., 2020). However, these systems typically start from a *well-specified structure* and are not designed to verify claims stated in natural language. PHYVER bridges this gap by mapping claims to structures and simulation protocols, and then aggregating MLIP- and DFT-based evidence into an interpretable verification output.

6 Conclusion

PHYVER emphasizes *knowledge-aware, physics-grounded* verification for material claims. Rather than relying on textual plausibility, it maps a natural-language claim into structured, physical-checkable predicates guided by domain knowledge (materials entity, conditions, and target properties), then instantiates an executable candidate and validates it with *multi-fidelity physical evidence*. In particular, MLIP-based relaxation offers fast feasibility and stability signals (e.g., convergence trends and residual forces), while optional DFT provides higher-confidence, property-level verification when needed; all evidence is surfaced through an interactive, step-wise interface for transparent inspection. Empirically, this knowledge-aware and physics-grounded verification improves agreement with expert judgment relative to a text-only LLM verifier.

Limitations and Broader Impact

PHYVER is constrained by the fidelity and runtime of its physics engines. On the fast side, UMA/MLIP optimization is efficient for screening and relaxation, but it can be unreliable for chemistries or bonding environments that are underrepresented in training data, and the resulting relaxed geometries (forces/energies) may be biased or overconfident for out-of-distribution structures. On the high-fidelity side, DFT provides stronger evidence but is the main runtime bottleneck: single-point and gradient calculations can take minutes to hours for large cells, defects, or complex compositions, limiting throughput and making interactive verification less responsive. Moreover, DFT outcomes depend on methodological choices (functional, basis/pseudopotentials, convergence), so errors or non-convergence can propagate into the final verdict, especially for small energy differences or comparative claims.

By translating free-form claims into executable checks and reporting quantitative evidence, PHYVER can improve scientific discovery by filtering physically implausible hypotheses early, prioritizing promising candidates for deeper simulation or experiments, and reducing fluent-but-ungrounded reasoning in scientific assistants. More broadly, PHYVER helps bridge the gap between experimental experts and the ML community: it exposes familiar physical signals (structures, relaxation behavior, computed properties) in an inspectable report, while providing ML practitioners with physics-grounded feedback loops that encourage reproducible, test-driven model development. As with any computational tool, PHYVER should be used to complement, not replace, careful expert judgment and experimental validation.

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A Appendix

A.1 Experiment Settings and Dataset Details

Implementation Details. PHYVER consists of four stages: (1) claim-to-structure generation, (2) MLIP-based optimization, (3) optional DFT computation, and (4) evidence-grounded claim verification.

For **generation**, the proposed disentangled diffusion model is revised from MetaSymbO (Chen et al., 2025) for adapting to chemical materials, checkpoint trained on the OMAT24 (Barroso-Luque et al., 2024) dataset. The LLM scaffold (default: GPT-5.1) is used only for entity extraction and structured prompting, while structure generation is from the disentangled diffusion model.

For **MLIP-based relaxation**, we use the FairChem Universal ML interatomic potential (UMA-S-1p1) checkpoint (uma-s-1p1.pt), accessed via ASE wrappers as described in the repository (Wood et al., 2025). We adopt the quick preset unless otherwise specified, which performs multi-loop anneal $\rightarrow$ quench $\rightarrow$ 0K relax cycles under GPU acceleration.

For **DFT computation**, we use ORCA (version 5/6 compatible) with the default configuration: M062X 6-31G* SP EnGrad D3BJ def2/J RIJCOSX TightSCF. Single-point energy and analytical gradients are computed on the MLIP-relaxed structure, and parsed outputs include total energy, dipole moment, HOMO/LUMO energies, and gap when available.

For **claim verification**, we collect all physical evidence, including MD optimization trajectories, DFT computed properties, and 3D geometries, and forward them to LLM for final verifications.

Training Data. The Disentangled Latent Diffusion generator is trained on the OMAT24 crystal dataset. The UMA model is pretrained on large-scale open catalyst and materials datasets as provided by FairChem. No additional fine-tuning is performed for the quantitative evaluation. LLM backbones are used in inference-only mode.

Evaluation Dataset. We evaluate 25 expert-created material claims spanning alloys, semiconductors, and battery systems. Each claim is annotated with a 5-point Likert score in $\{-2, -1, 0, 1, 2\}$ by domain experts. These gold labels are used exclusively for evaluation and were not used during training.

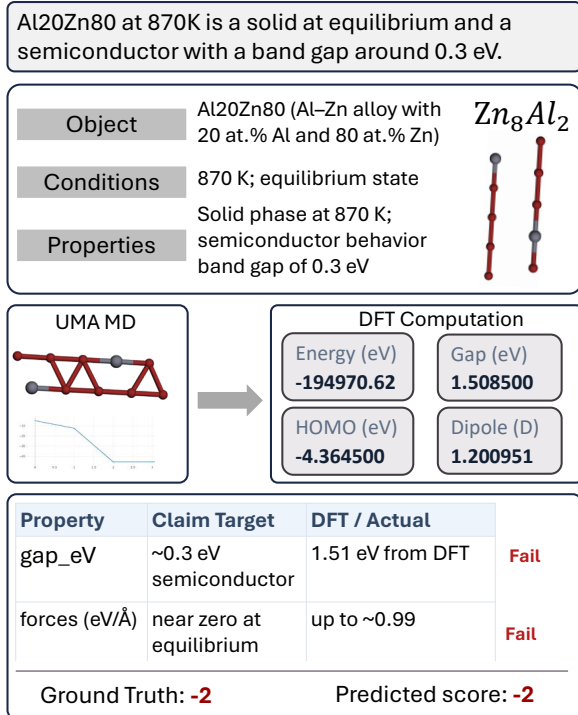


Figure 6: Case study. The predicted score is well matched with the expert-evaluated gold score.

Quantitative Evaluation Protocol. For each claim, we compare vanilla GPT-5.1 (text-only verification) and PHYVER-GPT-5.1 (LLM + structured physics evidence). We report Mean Absolute Error (MAE), Accuracy@ ± 1 (prediction within one Likert level), and Signed MAE to assess directional bias. All runs use the same LLM backbone and identical prompts for fairness; PHYVER additionally invokes structure generation, MLIP optimization, and optional DFT computation.

A.2 Additional Case Study

Figure 6 presents a failing example where PHYVER evaluates the claim *Al₂₀Zn₈₀ at 870K is a solid at equilibrium and a semiconductor with a band gap around 0.3 eV* and rejects it based on inconsistent structural stability and electronic evidence.

From this case study, we have the following observations: **First**, entity extraction decomposes the compound claim into multiple verifiable constraints: composition (Al₂₀Zn₈₀), thermodynamic condition (870K, equilibrium solid), and electronic property (band gap $\approx$ 0.3 eV), enabling structured and targeted verification rather than a holistic text-level judgment. **Second**, MLIP-based relaxation provides direct physical plausibility signals: the optimized structure exhibits large

residual forces (up to ~ 0.99 eV/Å), contradicting the assumption of equilibrium stability under the claimed condition. **Third**, DFT computation supplies decisive electronic evidence: the calculated band gap (~ 1.51 eV) significantly deviates from the claimed 0.3 eV, leading to a clear failure on the semiconducting constraint. By aggregating these independent physical signals, PHYVER outputs a predicted score of -2 , fully consistent with the expert-evaluated gold label (-2), demonstrating reliable physics-grounded rejection of an implausible material claim.

A.3 Prompts Used in PHYVER

We summarize the LLM prompts used in PHYVER and clarify how each prompt aligns with the system stages. We append the detailed prompt in the supplementary due to page limitations.

Material Instantiation (Step 0–1: Claim to Structure).

- **ENTITY_EXTRACTION** (*Claim parsing*): maps a free-form material claim into a structured *claim specification* consisting of *material object* (composition/prototype keywords), *conditions* (e.g., T , P), and *claimed properties* (what should be verified). This output serves as the executable interface for downstream material generation and physics-tool invocation.
- **INSTRUCTIONS_TRANSLATOR** (*Structure scaffolding*): converts the extracted claim specification into a minimal *generation scaffold* (e.g., FCC/rocksalt/perovskite) with a fixed schema (element types/atomic numbers and fractional coordinates). The scaffold provides a canonical starting point for the generator, improving stability and controllability of the instantiated candidates.

Evidence-Grounded Claim Verification (Step 4: Verdict from Physical Evidence).

- **SYSTEM_PROMPT** (*Evidence-grounded verification*): produces the final verdict by reasoning *only* over the provided structured evidence, including the instantiated/relaxed structure summary (Step 1–2 artifacts) and the parsed DFT outputs when available (Step 3). The prompt enforces explicit quantitative checks against claim predicates and outputs a calibrated Likert score with a structured JSON rationale.

Text-only Baseline (LLM-only Scoring Without Physics Evidence).

- **SYSTEM_PROMPT** (*Minimal Likert scoring*): returns a single Likert truthfulness score and a short JSON reason based only on the raw claim (and lightweight metadata), without access to PHYVER’s instantiated structures, optimization trajectories, or DFT evidence.